Invariants, Monovariants, Coloring & Combinatorial Games Practice Exercises

1 The integers $1, 2, 3, \ldots, 10$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 1$. After 9 steps, a single integer remains. What is that integer?

a. 45

b. 47

c. 46

d. 55

2 The integers $1, 2, 3, \ldots, 11$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 2$. After 10 steps, a single integer remains. What is that integer?

a. 44

b. 46

c. 48

d. 66

3 The integers $1, 2, 3, \ldots, 12$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 3$. After 11 steps, a single integer remains. What is that integer?

a. 45

b. 42

c. 48

d. 78

4 The integers $1, 2, 3, \ldots, 13$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 4$. After 12 steps, a single integer remains. What is that integer?

a. 39

b. 47

c. 91

d. 43

5 On a chalkboard are written the 8 fractions $\frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{9}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. 5

b. $\frac{8}{3}$

c. 4

d. $\frac{9}{2}$

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6 On a chalkboard are written the 9 fractions $\frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{11}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. 4

b. 3

c. $\frac{9}{4}$

d. $\frac{10}{3}$

7 On a chalkboard are written the 10 fractions $\frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{13}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. $\frac{5}{2}$

b. $\frac{7}{2}$

c. 2

d. $\frac{11}{4}$

8 On a chalkboard are written the 11 fractions $\frac{1}{5}, \frac{1}{6}, \ldots, \frac{1}{15}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. $\frac{16}{5}$

b. $\frac{11}{6}$

c. $\frac{12}{5}$

d. $\frac{11}{5}$

9 In normal-play Nim with three heaps of sizes $(3,\ 5,\ 7)$ (illustrated), a winning move exists on the third heap (size 7). How many stones must the first player **remove** from that heap of size 7 to leave a balanced Nim-sum 0 position?

a. 6

b. 3

c. 1

d. 2

10 In normal-play Nim with three heaps of sizes $(4,\ 7,\ 9)$ (illustrated), a winning move exists on the third heap (size 9). How many stones must the first player **remove** from that heap of size 9 to leave a balanced Nim-sum 0 position?

a. 3

b. 6

c. 10

d. 7

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11 In normal-play Nim with three heaps of sizes $(5,\ 8,\ 11)$ (illustrated), a winning move exists on the first heap (size 5). How many stones must the first player **remove** from that heap of size 5 to leave a balanced Nim-sum 0 position?

a. 2

b. 3

c. 6

d. -2

12 In normal-play Nim with three heaps of sizes $(6,\ 9,\ 13)$ (illustrated), a winning move exists on the first heap (size 6). How many stones must the first player **remove** from that heap of size 6 to leave a balanced Nim-sum 0 position?

a. 4

b. 3

c. -2

d. 2

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